# Supplementary Material: CL-TAV Contracts, Wrappers and Decision Tables

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**Abstract**

This supplement is part of the delivered manuscript package. It typesets ALG-CLTAV-02/03/04 wrappers, ALG-CLTAV-05/06/07 cores, typed records and the A/P stop tables. It does not change ALG-CLTAV-01 and does not report experiments. Supplement algorithm numbers are S1, S2, ...; they are not the main-text Algorithm 1.

## 1 Scope and numbering

Main-text Algorithm 1 is ALG-CLTAV-01. Supplement Algorithms S1–S6 are ALG-CLTAV-02, 03, 04, 05, 06 and 07. S9 in the main text calls CommitCompatibleUpdate (S6 / ALG-CLTAV-07), not a bare `IF-HIST-UPDATE` invocation.

## 2 S0–S10 duty map

| Step | Call | Duty / exception |
|---|---|---|
| S0 | Initialize | $\Gamma, \eta, H, U$; distinct $k$, charges, retry |
| S1 | `IF-RESOURCE-STOP` with `IF-EQUIV` | already decided Admit A2–A5 or P-stop |
| S2 | PredictCurrent | ALG-CLTAV-02; TEST-scoped GAP |
| S3 | SelectAndAdmit | ALG-CLTAV-05; freeze $\xi$ including $U$ |
| S4 | ChargeOnce | one charge; retry cap $\neq$ budget |
| S5 | `IF-EXECUTE-RECORD` | raw record and effect class |
| S6 | InterpretOutcome | ALG-CLTAV-03 + timing ALG-CLTAV-06 |
| S7–S8 | ResolveOutcome | ALG-CLTAV-04; adopt returned contexts |
| S9 | CommitCompatibleUpdate | version/intersection; SPEC-ERROR; summary commit |
| S10 | `IF-RESOURCE-STOP` with `IF-EQUIV` | P-stop or continue to S1 |

## 3 Typed records

| Record | Fields |
|---|---|
| PredictionResult | status OK\|GAP; reason; classesByTest (OK); historyVersionUsed; uncertaintyRef |
| Decision | kind ACTION\|EXIT\|SPEC-ERROR; actionKind; actionId; classesUsed; reason |
| SelectSnapshot $\xi$ | actionId; actionKind; qUsedAtSelect; historyVersion; classesUsedAtSelect; uncertaintyRef |
| ExecutionResult | attemptId; record; effect; correlationId |
| Outcome $z$ | validity; effect; kind; actionId; $I_z$ or NONE; prepResultEvaluated; targetConfirmed; summaryConfirmed; prepErr; declaredTarget; evidence; postSummary; timing |
| Resolution $\delta$ | control COMMIT\|RETRY\|STOP; $\Gamma'$; $\eta'$; outcome; reason |

# 4 Admit and post-update stops

A1 selects one affordable eligible action from the exclusive TEST, Prep or Recover alternatives; distinguishing value is required only for TEST. A2 is KNOWN with a distinguishing TEST or eligible Prep, but no such action affordable (Stop-Budget). A3 is KNOWN with no distinguishing TEST and no eligible Prep (Stop-NoDistinguisher). A4 is UNKNOWN with no eligible Recover (Stop-Error). A5 is UNKNOWN with Recover eligible but unaffordable (Stop-Budget). A TEST prediction GAP is handled independently: a KNOWN GAP without eligible Prep is SPEC-ERROR, not A3; retry-cap remains an independent Stop-Error condition. P1 Stop-Empty; P2 Stop-Singleton; P3 Stop-645; P4 Stop-Equivalent; P5 Stop-Budget after update. Stop-NoDistinguisher is exclusive with those P-stops when no remaining distinguisher exists and no P-stop already fired. Order is Empty, Singleton, 645, Equivalent, Budget. Singleton $h_{\text{normal}}$ is not protocol PASS.

# 5 Short arguments

Monotonicity, retention (R1)–(R5), one-step minimax and resource termination are stated in the main text. The algebraic details are the accepted method §3.9.1 arguments: update is intersection; `ERROR` does not exclude; minimax is a least element of a finite selectable set; billed executions are bounded by the selected mode. They are not global optimality and not independent mathematical approval.

# 6 Experiment-design accounting contract

The candidate design reuses four arms, three families and eight registered scenario identifiers. One unique `attemptId` is the all-ledger unit; `runId` exists only after admission. Truth joins arm output only at evaluation. Detection answered, localization decision, truth-scorable subsets, explicit missing-truth flow, true-hypothesis retention, singleton correctness, abstention, error, equivalence, unresolved fault and budget exit remain distinct, potentially overlapping labels. Every result cell names its finite population, missing/censoring rule and metric-specific empty-denominator behavior. With registered attempts but no answers, answered coverage is (0/N) while answered-conditional error is N/A. Recovery detail is folded into preparation exactly once in the accepted four-component cost projection. Pre-run review/setup effort is reported separately from issued-action cost. An update rejected as `ERROR` preserves the preceding hypothesis set regardless of its stop tag; only a valid empty compatibility update produces `Stop-Empty`, and a valid singleton update first produces `Stop-Singleton`. A registered pair is derived separately for each parent metric and requires the same base scene, seed, construction and registered contrast, reciprocal arms, and distinct confirmed resets. The closed-interval T5 rule assigns containment PASS, disjointness FAIL and partial overlap INCONCLUSIVE. Confirmatory execution is blocked until the IUT/reset, truth/injection, equivalence, mathematical-review, sample/repeat, resource-budget and clock/error budget decisions are frozen. The machine-checked synthetic ledger is an accounting witness, not evidence of method performance.

# 7 AI disclosure fields

System: PENDING-USER (AUTH-AI). Scope (text/figures/code): PENDING-USER. Author verification: required; not completed by this sentence. IEEE AESS guidance is rechecked at submission.

## 8 Wrappers and core mechanisms

**Algorithm S1:** PredictCurrent: current-observation prediction wrapper (ALG-CLTAV-02, S2)

**Require:** SessionContext $\Gamma$; HistoryHandle $\eta$; current $H$; uncertainty reference $U$; action library $L$

**Ensure:** PredictionResult: status OK with nonempty classesByTest, or status GAP with a reason; the session's own $\eta$ is read, not rebuilt

$raw \leftarrow$ `IF-PRED-OBS`$(\Gamma, \eta, H, U, L)$;

`// branch on the backend tag BEFORE reading or projecting classes; a GAP result carries no classesByTest`

**if** $raw$.status = *GAP* **then**

  **return** PredictionResult(status = `GAP`, reason = $raw$.reason);

  `// the gap stays TEST-scoped`

$classes \leftarrow$ ProjectOntoCurrentH($raw$.classes, $H$);

`// a declared class set that projects to no nonempty class is the same named gap, not OK and not` $score = 0$

**if** *classes has no nonempty member* **then**

  **return** PredictionResult(status = `GAP`, reason = no currently valid class);

**return** PredictionResult(status = `OK`, classesByTest = *classes*, historyVersionUsed = $\eta$.version, uncertaintyRef = $U$);

**Algorithm S2:** InterpretOutcome: execution and observation coordination (ALG-CLTAV-03, S6)

**Input:** ExecutionResult $er$; SelectSnapshot $\xi$; $\Gamma, \eta$; uncertainty reference $U$
**Output:** total Outcome $z$, including effect, validity, $I_z$, and confirmation fields

$z \leftarrow$ NewOutcome(); $z$.effect $\leftarrow$ $er$.effect; $z$.kind $\leftarrow$ $\xi$.actionKind; $z$.actionId $\leftarrow$ $\xi$.actionId;
**if** $er$.effect $\in$ {*CONFIRMED-NOT-SENT*, *UNKNOWN-EFFECT*} **then**
    **return** $z$;
$(I_z, e, s, p, o, m) \leftarrow$ `IF-OBS-INTERPRET`$(\Gamma, er.\text{record}, \text{clocks}, \varepsilon)$;
**if** $o$ *is invalid* **then**
    set $z$.validity $\leftarrow$ ERROR and $z$.effect $\leftarrow$ `UNKNOWN-EFFECT`; **return** $z$;
timing $\leftarrow$ InterpretTimedObservation$(o, m, \xi, N_r, D_r, \text{Match}_r)$;
**if** timing.verdict = ERROR **then**
    set $z$.validity $\leftarrow$ ERROR and $z$.effect $\leftarrow$ `UNKNOWN-EFFECT`; **return** $z$;
set $z$.validity $\leftarrow$ timing.verdict and $z.I_z \leftarrow I_z$;
**if** $\xi$.actionKind $\in$ {PREP, RECOVER} **then**
    `// IF-PREP-RECOVER returns values only; it does not write Γ or η`
    $z$.prepResultEvaluated $\leftarrow$ **true**;
    $(z.\text{targetConfirmed}, s, z.\text{prepErr}, z.\text{declaredTarget}, z.\text{evidence}, p) \leftarrow$ `IF-PREP-RECOVER`$(\Gamma, \eta, \xi.\text{actionId}, er.\text{record})$
set $z$.summaryConfirmed $\leftarrow$ $s$ and $z$.postSummary $\leftarrow$ $p$;
**if** $e =$ *UNKNOWN-EFFECT*, $z$.prepErr, *an unconfirmed Prep summary, an unconfirmed Recover target, or* $e =$ *NONE with no confirmed summary* **then**
    set $z$.effect $\leftarrow$ `UNKNOWN-EFFECT`;
**else**
    set $z$.effect $\leftarrow$ $e$;
`// all Γ/η writes stay in ALG-CLTAV-04 and S9`
**return** $z$;

**Algorithm S3:** ResolveOutcome: ERROR, retry and confirmed recovery (ALG-CLTAV-04, S7–S8)

**Require:** Outcome $z$ (effect, kind, validity, confirmation); SelectSnapshot $\xi$; $\Gamma$; $\eta$
**Ensure:** Resolution $\delta = (\text{control}, \Gamma', \eta', \text{outcome}, \text{reason})$: $\Gamma'$ initialized from $\Gamma$

$\Gamma' \leftarrow \Gamma$; $\eta' \leftarrow \eta$;
`// copy/persist semantics: the returned contexts are total, never undefined`
**if** $z$.validity = ERROR *or* $z$.effect $\in \{$*CONFIRMED-NOT-SENT*, *UNKNOWN-EFFECT*$\}$ *or* $z$.prepErr *or* (*z*.kind *is Prep and* *z*.prepResultEvaluated *and* *z*.summaryConfirmed *is false) or* (*z*.kind *is Recover and* *z*.prepResultEvaluated *and* *z*.targetConfirmed *is false)* **then**
  `// S7 classify by kind and evaluated confirmation; do not fold CONFIRMED-NOT-SENT into UNKNOWN`
  do not exclude $H$;
  **if** $z$.validity = ERROR *or* $z$.prepErr *or* $z$.effect = *UNKNOWN-EFFECT* *or* (*z*.kind *is Prep and* *z*.prepResultEvaluated *and* *z*.summaryConfirmed *is false)* **then**
    mark $\eta'$ conservative-unknown; $\Gamma'$.qStatus $\leftarrow$ `UNKNOWN`; clear $\Gamma'$.currentSummary;
  **else if** $z$.effect = *CONFIRMED-NOT-SENT* **then**
    leave $\Gamma'$.qStatus and compatible history unchanged;
  **else**
    $\Gamma'$.qStatus stays UNKNOWN; clear $\Gamma'$.currentSummary; mark $\eta'$ conservative-unknown;
    `// unconfirmed Recover target`
  $\Gamma'$.retryCount $\leftarrow \Gamma$.retryCount + 1;
  **if** $\Gamma'$.retryCount $\geq \Gamma'$.retryCap **then return** Resolution(control = `STOP`, $\Gamma', \eta', z$, reason = Stop-Error);
  **return** Resolution(control = `RETRY`, $\Gamma', \eta', z$);
  `// an ERROR/unknown outcome reaches retry, never commit`
**else if** $z$.kind *is Recover* **then**
  `// S8 a confirmed Recover is a necessary condition for this branch; sole writer of KNOWN; the normalized summary returns to S9`
  $\Gamma'$.qStatus $\leftarrow$ `KNOWN`; $z$.postSummary $\leftarrow$ ($z$.declaredTarget, $z$.evidence);
**return** Resolution(control = `COMMIT`, $\Gamma', \eta', z$);
`// a normal TEST/Prep with a valid observation commits; the normalized Outcome carries the confirmed postSummary to S9`

**Algorithm S4:** SELECTANDADMIT: test selection and resource admission (ALG-CLTAV-05, S3)

**Require:** $\Gamma$; $\eta$; current $H$; action library $L$; typed *pred* (PredictionResult)
**Ensure:** a Decision: kind ACTION / EXIT / SPEC-ERROR with actionKind, actionId, classesUsed and reason

`// state eligibility is independent of affordability; under UNKNOWN only an eligible Recover qualifies, never a Prep`

eligible$(t) \leftarrow (\Gamma.\text{qStatus} = \texttt{UNKNOWN}\ ?\ (t.\text{kind} = \text{RECOVER} \wedge t.\text{eligibleUnknown})\ :\ (t.\text{kind} \neq \text{RECOVER} \wedge t.\text{enabledAt} = \Gamma.\text{currentSummary}))$;
$A \leftarrow \{t \in L : \text{eligible}(t)\}$;

`// affordability is decided by IF-SELECT-ADMIT, not assumed from eligibility`
`// a global or per-TEST projection gap withholds TEST scoring and keeps the gap identity; it does not itself block eligible Prep/Recover`

*testGap* $\leftarrow$ **false**;
**if** *pred*.status = *GAP* **then**
  $C \leftarrow \{\}$; *testGap* $\leftarrow$ **true**;
**else**
  $C \leftarrow \{\mathcal{C}_t\}$ with $\mathcal{C}_t \leftarrow \{H \cap C : C \in \textit{pred}.\text{classesByTest}[t],\ H \cap C \neq \varnothing\}$ for each TEST $t \in A$;
  **if** *a TEST* $t \in A$ *has declared classes but* $\mathcal{C}_t = \varnothing$ **then**
    $C \leftarrow \{\}$;
    *testGap* $\leftarrow$ **true**;

`// IF-SELECT-ADMIT is the single admission authority: affordability and A1-A5; exclusive kind; one-step minimax` $s(t) = \max_{C \in \mathcal{C}_t} |C|$ `over currently valid nonempty classes of affordable distinguishing TESTs` $D \cap F$ `only; cost/id tie-break; never` $\arg\min \varnothing$`; Prep/Recover are not observation minimax; uninformative TESTs are not A2`

$(k, S, t^\star, \textit{admit}) \leftarrow \texttt{IF-SELECT-ADMIT}(A, \Gamma.\text{currentSummary}, C, \Gamma, \eta, H)$;
**if** $k$ = ADMIT **then**
  **if** *testGap and admit* = A3 **then return** Decision(kind = `SPEC-ERROR`, reason = *pred*.reason or per-test prediction gap);
  **return** Decision(kind = `EXIT`, reason = the A2–A5 admission reason);
**else if** $k$ = PREP **then**
  **return** Decision(kind = `ACTION`, actionKind = `PREP`, actionId = $t^\star$, classesUsed = $S$);
**else if** $k$ = RECOVER **then**
  **return** Decision(kind = `ACTION`, actionKind = `RECOVER`, actionId = $t^\star$, $\cdot$);
**else if** *testGap* **then**
  **return** Decision(kind = `SPEC-ERROR`, reason = *pred*.reason);
**return** Decision(kind = `ACTION`, actionKind = `TEST`, actionId = $t^\star$, classesUsed = $\mathcal{C}_{t^\star}$);

**Algorithm S5:** InterpretTimedObservation: timed observation kernel (ALG-CLTAV-06, S6 kernel)

**Require:** the single ownership result and the raw measurement interval from IF-OBS-INTERPRET; SelectSnapshot $\xi$; named obligation inputs $N_r$ (allowed interval) and $D_r$ (physical domain); declared match policy $\text{Match}_r$; declared uncertainty $\varepsilon$

**Ensure:** a TimingResult: verdict in {PASS, FAIL, INCONCLUSIVE, ERROR}; the single pairing evidence; the interval used; validity

`// resolve` $N_r$, $D_r$ `and` $\varepsilon$ `from the named obligation and instance inputs, not from free symbols`

pair $\leftarrow \text{Match}_r(\text{ownership}, \xi)$;

`// one pairing result feeds T2 and the no-response rule`

**if** pair *is ambiguous* **then return** TimingResult(verdict = ERROR);

**if** pair *is a legitimate trigger/response pair* **then**

  $M \leftarrow \text{measurementInterval} \cap D_r$;

  **if** $M = \varnothing$ **then return** TimingResult(verdict = ERROR);

  **if** $M \subseteq N_r$ **then**

    **return** TimingResult(verdict = PASS, evidence = pair, $M$);

  **else if** $M \cap N_r = \varnothing$ **then**

    **return** TimingResult(verdict = FAIL, evidence = pair, $M$);

  **return** TimingResult(verdict = INCONCLUSIVE, evidence = pair, $M$);

  `// open/closed endpoints preserved`

`// a missing response forms a no-response only after a still-active instance reaches its deadline; this round not seeing it is not a timeout`

**if** *the obligation is still active and expired* **then**

  `// use the earliest admissible elapsed horizon (a scalar), with the requirement's boundary convention`

  **if** $N_r$ *is closed at* $U_r$ **then**

    expired $\leftarrow$ lowerHorizon $> U_r$;

  **else**

    expired $\leftarrow$ lowerHorizon $\geq U_r$;

  **if** expired **then return** TimingResult(verdict = FAIL);

  **return** TimingResult(verdict = INCONCLUSIVE);

**return** TimingResult(verdict = ERROR);

`// a missing pair is not a point observation`

**Algorithm S6:** CommitCompatibleUpdate: compatible update and summary commit (S6 / ALG-CLTAV-07)

**Input:** snapshot $\xi$; Outcome $z$; contexts $\Gamma, \eta, H$
**Output:** UpdateResult $(status, \eta', H', \Gamma')$

$\Gamma' \leftarrow \Gamma$;
**if** $\xi$.historyVersion $\neq \eta$.version **then return** SPEC-ERROR with unchanged $\eta, H, \Gamma'$;
**if** $z.I_z$ *is a valid compatible class* **then**
    $(\eta_c, H_c) \leftarrow$ `IF-HIST-UPDATE` with:;
        $\eta, H, \xi$.actionId, $\xi$.qUsedAtSelect, $\xi$.classesUsedAtSelect,;
        $\xi$.historyVersion, $z.I_z$, $z$.postSummary;
    **if** $H_c \neq H \cap z.I_z$ **then return** SPEC-ERROR (intersection mismatch);
    $\eta' \leftarrow \eta_c$; $H' \leftarrow H_c$;
**else**
    $(\eta', H') \leftarrow$ `IF-HIST-UPDATE` with the same snapshot fields, NONE, and $z$.postSummary;
    $H' \leftarrow H$;
**if** $\Gamma'$.qStatus *is KNOWN and* $z$.summaryConfirmed **then**
  $\Gamma'$.currentSummary $\leftarrow z$.postSummary;
**return** OK with $(\eta', H', \Gamma')$;